# F.O.R.G.E.: Fine-tuned Optical Reconstruction for Gesture Extraction

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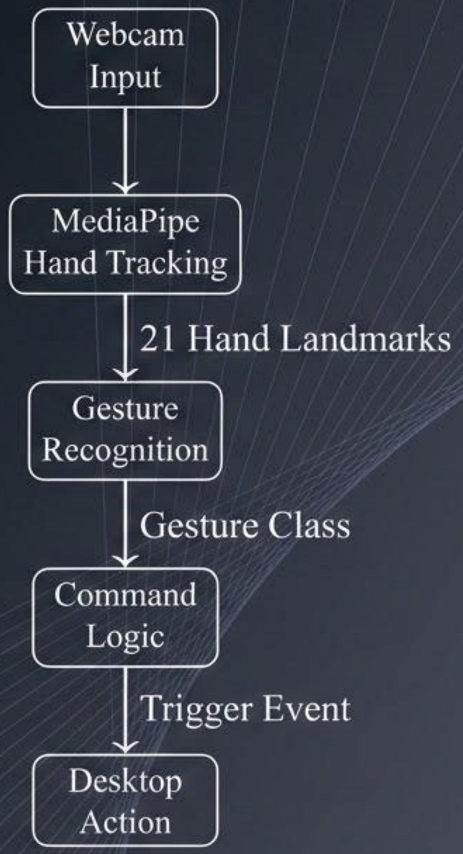
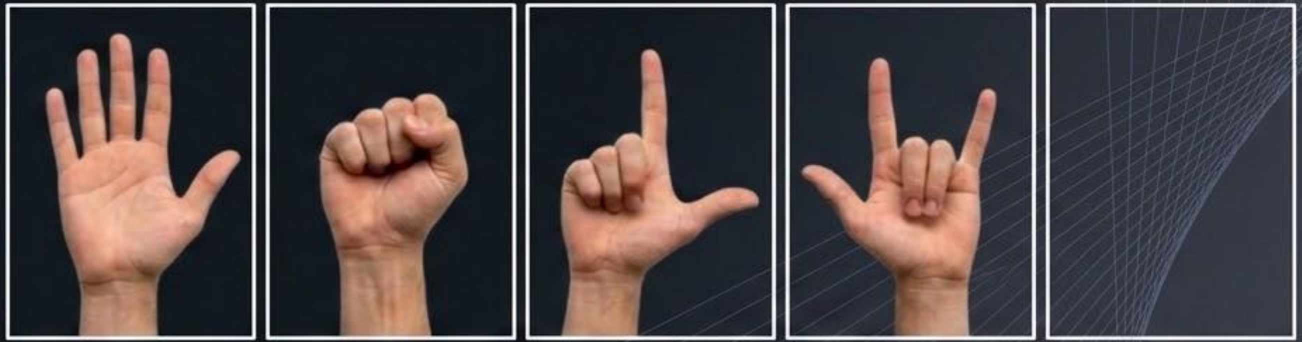
# Project Overview

- A custom hand gesture interaction system
- Convert gestures into user commands
  - Toggle a desk lamp
  - Switch desktops
  - Open gemini in a browser



# Data Collection

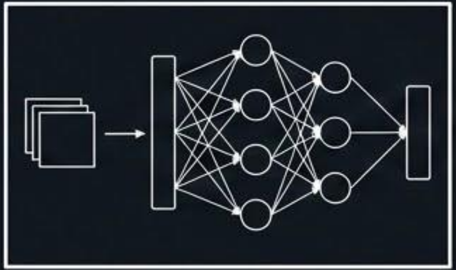
- Brainstormed meaningful gestures
- Collected our own data (~1000 images total)



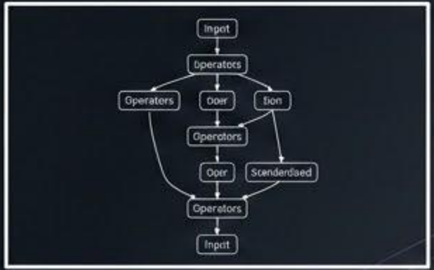
OPEN PALM | CLOSED FIST | L SHAPE | ILOVEYOU GESTURE | NO HAND

# Training & Optimization

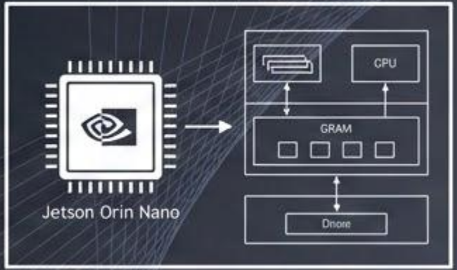
- 80/10/10 training/validation/test split for 10 epochs
- Increased model performance with TensorRT optimizations



Standard Network
(Mediapipe)

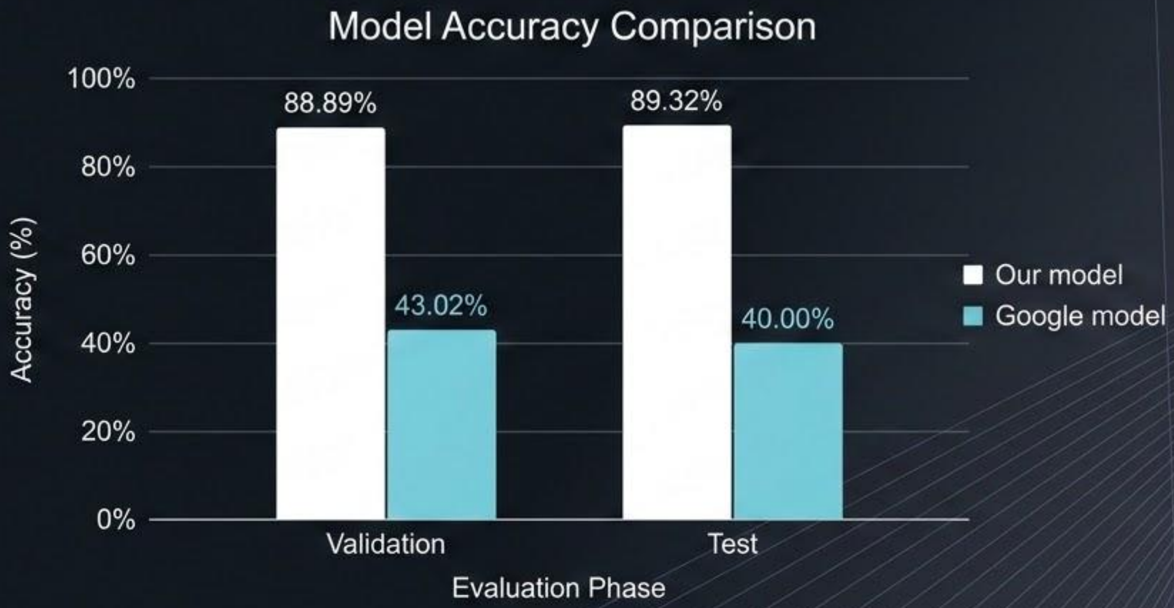


ONNX Format




ENGINE Format
(Jetson Orin Nano)

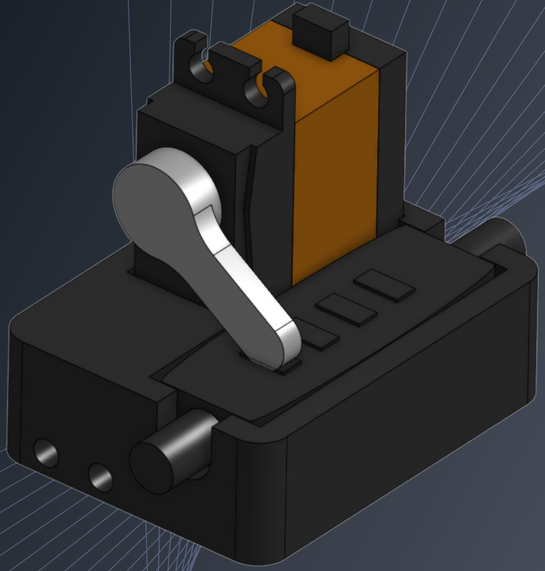
# Evaluation



Model Accuracy Comparison
100%
80%
60%
40%
20%
0%
Accuracy (%)
88.89%
43.02%
89.32%
40.00%
Validation
Test
Evaluation Phase
Our model
Google model

# Actions

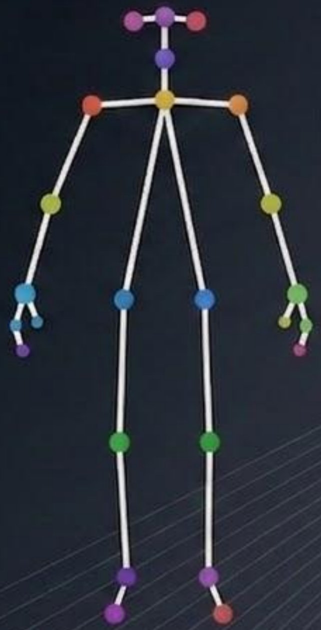
- Desk Lamp
  - Short pulse command sent over USB
  - Arduino Uno + 6V battery pack + torque servo + 3D printed mount
- Switch Desktops
  - Sends a system hotkey
- Open Gemini
  - Spawns Firefox with a custom URL



3D printed mount for servo + lamp controller

# Future Work

- Expand to full body tracking
- Implement LIDAR or similar light-independent gesture detection



Full body tracking

LIDAR gesture detection

# Demo